

MASS-ENG-003

Identity Engine Specification

Page 1 – Purpose & Scope

Purpose

Provide enterprise-wide identity, authentication, authorization, and principal management for every subsystem.

Objectives

- Establish a single identity authority
- Support users, service accounts, and system identities
- Centralize authentication and authorization
- Provide identity services to all platform engines

Out of Scope

Business-specific permissions and workflow rules.

Page 2 – Architecture

Core Components

Identity Store, Authentication Service, Authorization Service, Role Manager, Session Manager, Token Service.

Relationships

All platform engines consume Identity services through Enterprise Core.

Page 3 – Functional Specification

Requirements

1. Maintain unique enterprise identities.
2. Authenticate all principals.
3. Issue secure session/token credentials.
4. Evaluate authorization requests.
5. Record authentication events.

Non-Functional

Least privilege, auditability, extensibility, high availability.

Page 4 – Interfaces & Examples

Interfaces

Authenticate, Authorize, Identity Lookup, Role Assignment, Token Validation.

Example Flow

User → Authenticate → Identity Verified → Authorization Check → Access Granted/Denied → Audit Event.

Page 5 – Construction Package

Claude Checklist

Implement identity model; authentication service; authorization service; role management; session/token management; audit logging; automated tests.

Definition of Done

Authenticated principals can securely access authorized resources; all identity actions are auditable.

Constitution References

Identity, Security, Governance, Enterprise Structure.